

Neuromorphic ASR

Transducer Models on Memristor Hardware

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In-Memory Computing [Rossenbach & Hilmes⁺ 25]

- ▶ Various technologies exist, with varying tradeoffs:
 - ▷ Chip complexity and density
 - ▷ Energy consumption
 - ▷ Speed and reliability
- ▶ This talk focuses on **memristive** hardware, in particular **resistive RAM (ReRAM)**:
 - ▷ Key Benefits: low energy consumption, non volatility

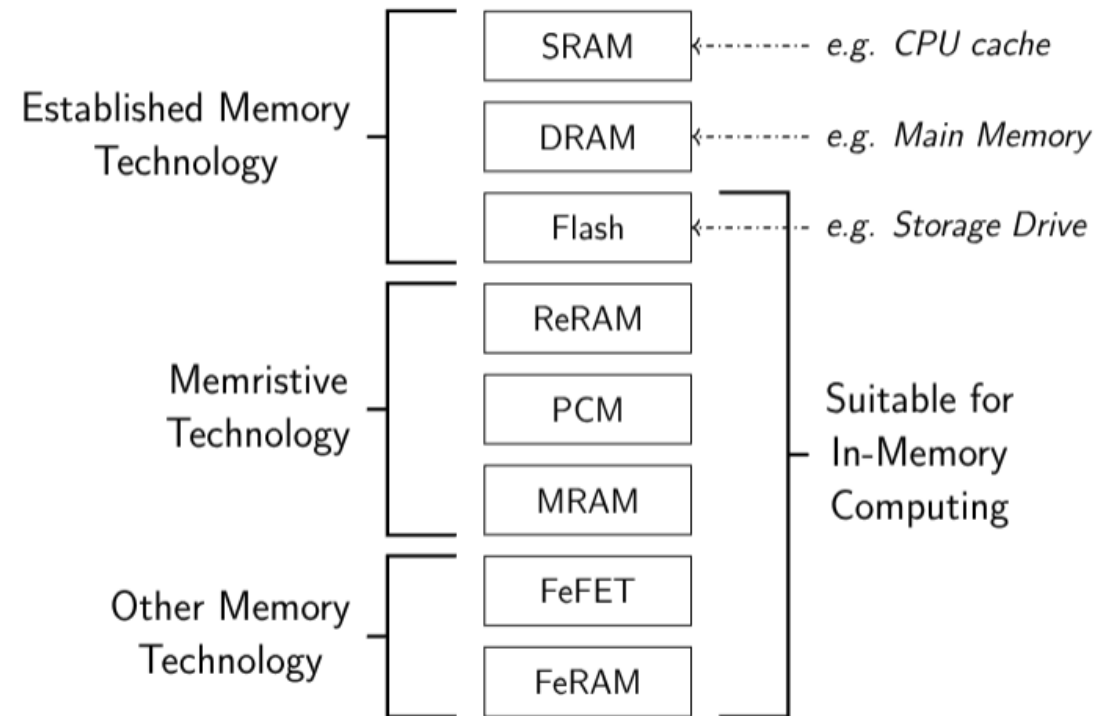


Figure: Taxonomy of In-Memory Computing technologies

Overview [Rossenbach & Hilmes⁺ 25]

- ▶ A **memristor** is an analog computing device with a programmable resistance state [Strukov & Snider⁺ 08].
- ▶ Enables fast, energy-efficient implementation of Vector-Matrix Multiplication (VMM) [Wan & Kubendran⁺ 22] -
→ and thus neural networks.
- ▶ Limitations:
 - ▷ High analog inaccuracies, and thus need specific restrictions on mathematical operations [Wouters & Chen⁺ 16].
 - ▷ Production is still in early stages [Infineon Technologies AG 22].
 - ▷ Current experiments are simulated using SynaptogenML [Hennen & Brackmann⁺ 24] [Rossenbach & Hilmes⁺ 25].

Resistor:



Memristor:

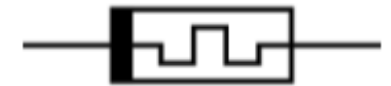


Figure: Resistor and Memristor schematics

Analog In Memory Computing [Aguirre & Sebastian⁺ 24]

- Multiplication computation done via Ohm's Law:

$$V = IR \quad (1)$$

- Voltage V , current I and resistance R .
- Possible to rewrite resistance as conductance G :

$$I = GV \quad \text{with} \quad G = \frac{1}{R} \quad (2)$$

- Addition done via Kirchhoff's Law for any connected node in a circuit:

$$\sum_{n=0}^N I_n = 0 \rightarrow I_0 = - \sum_{n=1}^N I_n \quad (3)$$

- Currents can be trivially added by simply connecting electric lines.

Analog In Memory Computing [Rossenbach & Hilmes⁺ 25]

- ▶ Resistors ($\square$) as connecting element between vertical and horizontal circuit lines
- ▶ Each resistor's conductance is an element of a constant matrix
- ▶ Column sum simply via line connection

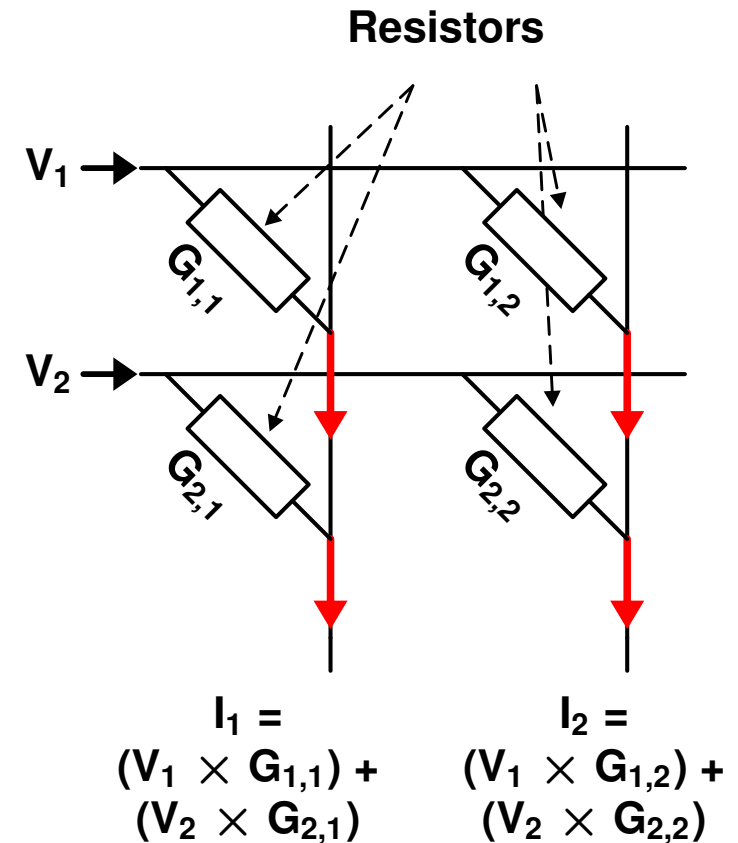


Figure: Crossbar array performing analog VMM [Rossenbach & Hilmes⁺ 25]

The Memristor [Hennen & Brackmann⁺ 24]

Key Idea: Program weights by setting the resistance.

- ▶ Apply $-V > -V_S \rightarrow$ set to low resistance
- ▶ Apply $V > V_R \rightarrow$ set to high resistance
- ▶ Intermediate states possible, but difficult to program reliably
- ▶ State is physically stored (non-volatile memory)

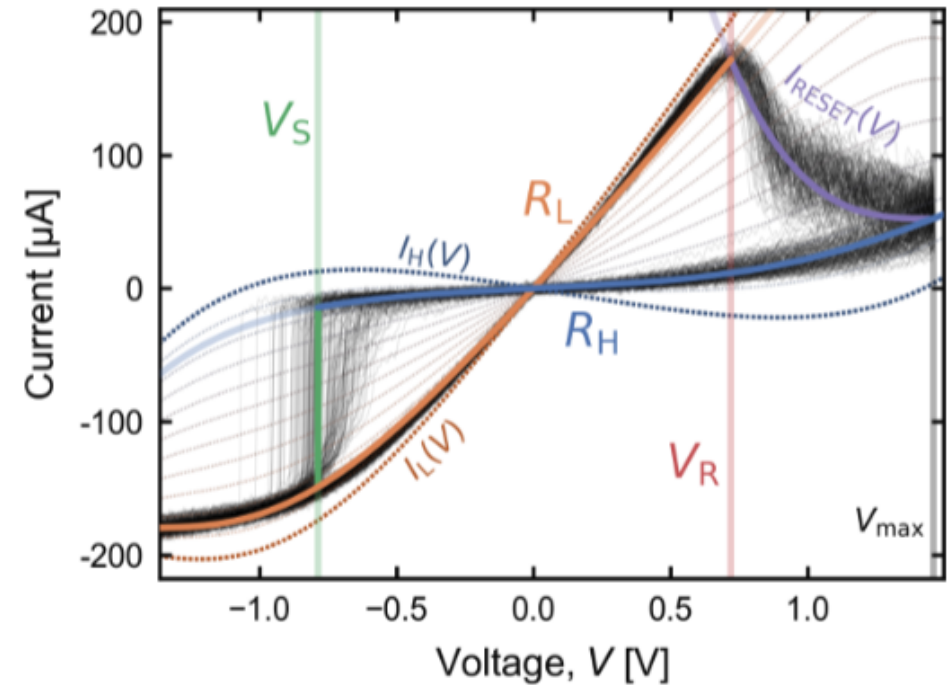


Figure: I-V characteristic curve of a memristor [Hennen & Brackmann⁺ 24]

Limitations [Rossenbach & Hilmes⁺ 25]

Programming Limitations

- ▶ Resulting resistance state varies with each programming cycle
- ▶ Non-linear relation of current and voltage

Cannot exactly model weights

- ▶ Only positive, non-zero weights possible
- ▶ Current leakage prohibits a true zero

(Paired) Memristor Array [Aguirre & Sebastian⁺ 24]

- ▶ Use positive and negative bitlines to create memristor pairs
- ▶ Each row-wise pair of memristors represents a weight
- Differential pair allows for negative and zero weight values:

$$I_j = \sum_{i=1}^N V_i G_{i,j}^+ - \sum_{i=1}^N V_i G_{i,j}^-$$

Problems:

- ▶ Cannot set paired memristors to identical conductances
- Difference close to zero, but not guaranteed

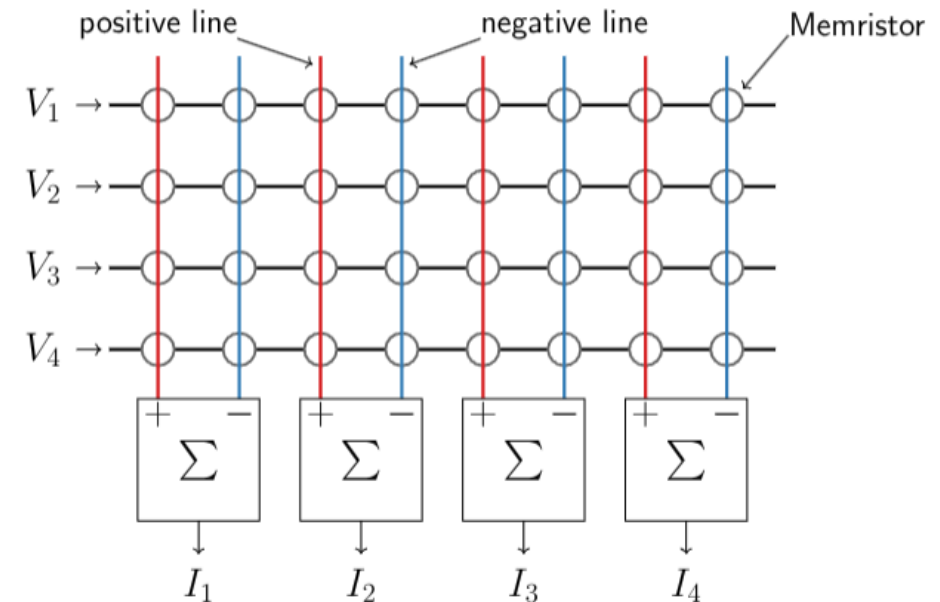


Figure: Paired memristor array structure [Aguirre & Sebastian⁺ 24]

Memristor Neural Layers [Rossenbach & Hilmes⁺ 25]

- ▶ **Tile** the weight matrix:
 - ▷ Tile weight matrix with arbitrary dimensions into smaller fixed-size arrays.
 - ▷ E.g., 512×512 weight matrix $\rightarrow$ 16 crossbar arrays of size 128×128
- ▶ **Slice** weights into memristor states:
 - ▷ Binary setting (high vs. low resistance)
 - ▷ Stacking of memristors for each magnitude bit (k -bit weight requires $k - 1$ arrays)
- ▶ **Pair** positive (+) and negative (−) bitlines.

Memristor Neural Layers

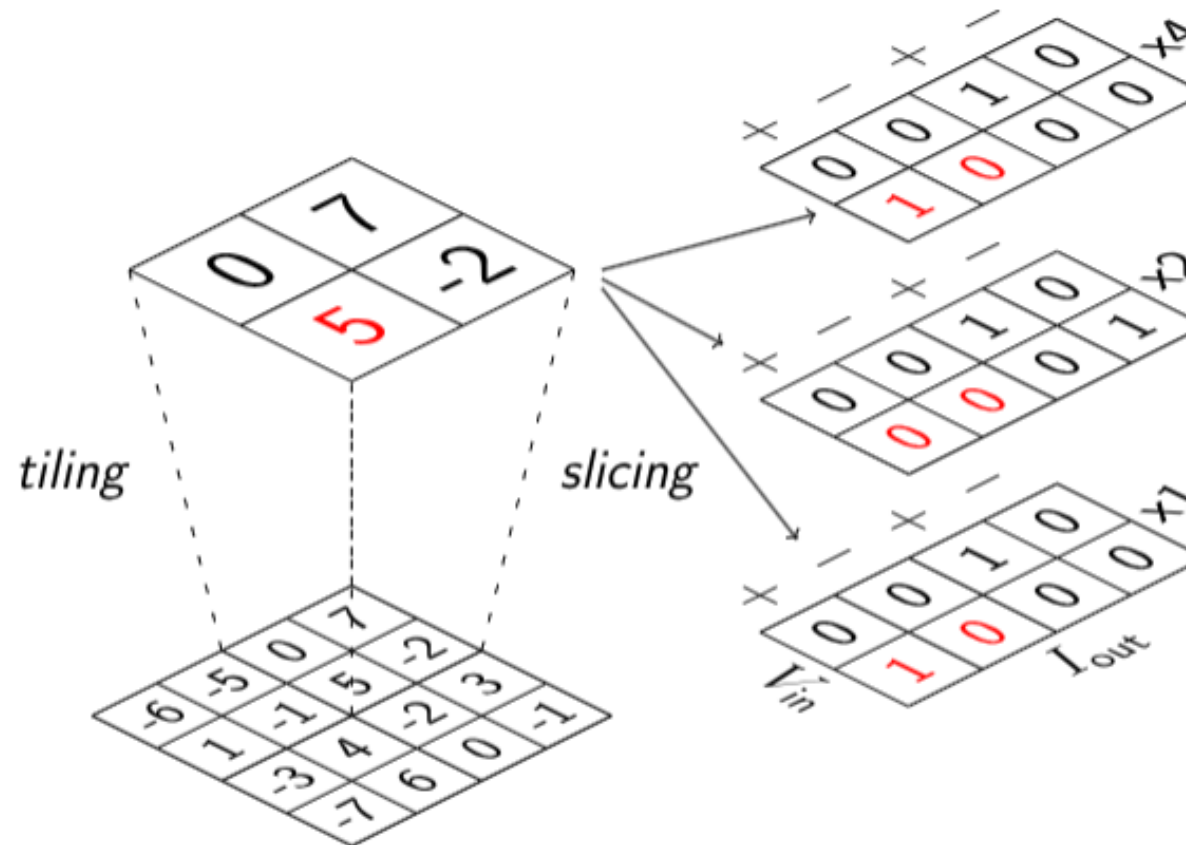


Figure: Mapping a weight matrix on memristor arrays [Rossenbach & Hilmes⁺ 25]

Quantization

- ▶ The number of memristors required is proportional to the precision of the weights.
 - ▷ Need to reduce the precision of the weights → quantization.
- ▶ Quantization-Aware Training (QAT) performs better than Post-training Quantization (PTQ), but requires more training time [Gholami & Kim⁺ 21].
 - High performance gap at low precision

Table: WER (%) on TED-LIUMv2 dev, Conformer CTC, 4-gram LM [Rossenbach & Hilmes⁺ 25]

Weight bits	FP32	8	6	5	4	3	2
PTQ	7.2	7.2	7.9	8.9	11.4	30.7	–
QAT		7.3	7.7	7.4	7.8	8.3	22.1

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Motivation

- ▶ CTC-based Conformer shown to run on (simulated) memristor hardware [Rossenbach & Hilmes⁺ 25].
- ▶ How do **transducer** models, with their added complexity compared to CTC, behave on device?
 - ▷ Autoregressive prediction network; internal language model (ILM).
 - ▷ Joint network evaluated repeatedly per emitted label.
- ▶ Performance degradation across search paradigms -
 - ▷ Lexicon-free vs. tree search.
 - ▷ ILM and prior scaling - How does device noise change the optimal LM weighting?

Motivation

- ▶ Performance impact of deploying components of the model on the device?
 - ▷ Encoder only vs. encoder + prediction network vs. full model (encoder + prediction network + joint network)
- ▶ **Tradeoff**: more offloading $\Rightarrow$ more energy efficiency, but also more noise.

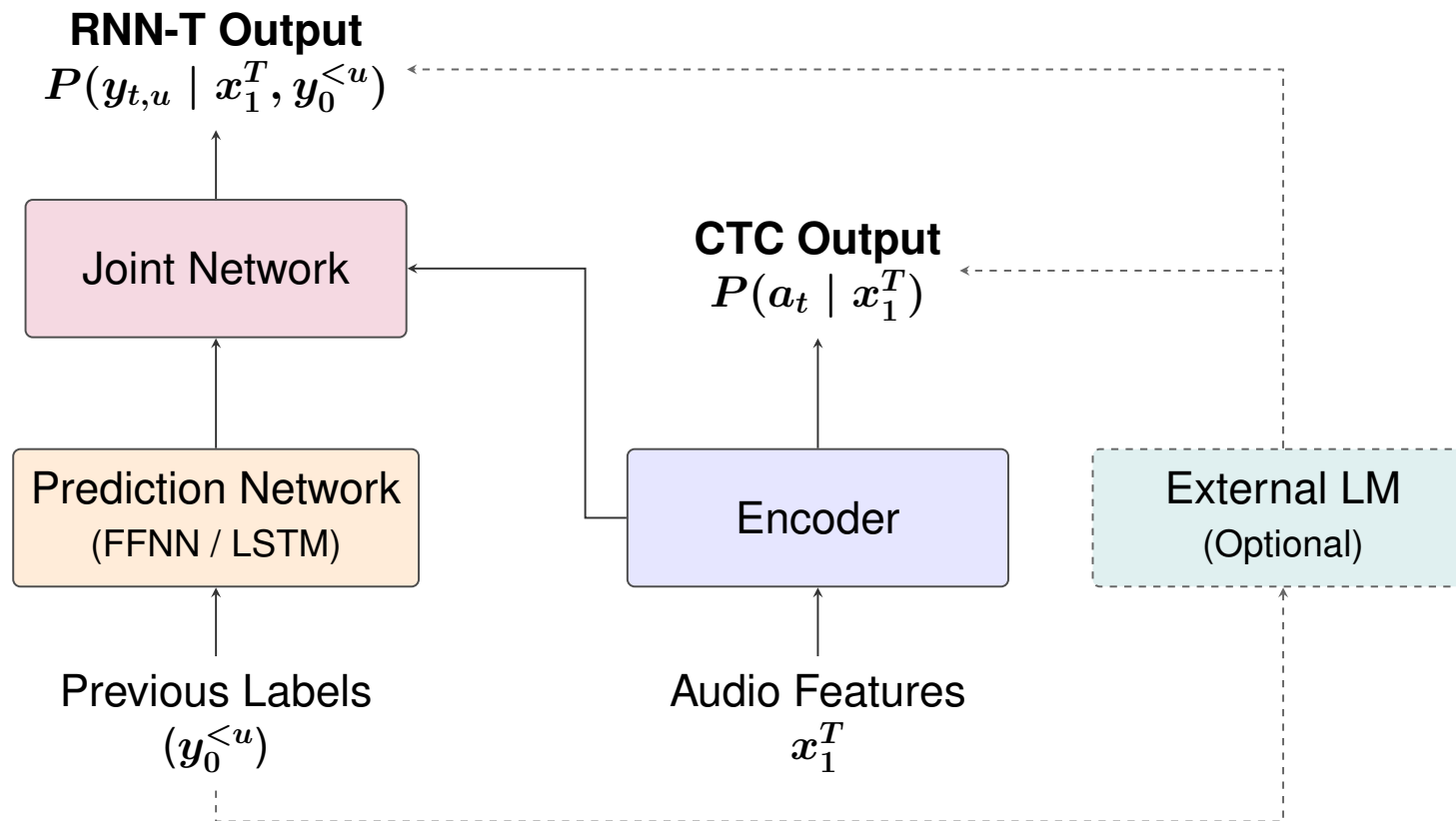


Figure: CTC vs. Transducer architecture components

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Experimental Setup

- ▶ Data: LibriSpeech [Panayotov & Chen⁺ 15] (960 h), BPE subword targets (128 merges → 184 labels).
- ▶ Features: 80 log-mel filters, 10 ms frame shift; SpecAugment during training.
- ▶ Encoder: 12-layer Conformer [Gulati & Qin⁺ 20], 512-dim.
- ▶ Model types:
 1. **CTC** [Graves & Fernández⁺ 06]: encoder only.
 2. **FFNN-transducer** [Graves 12]: prediction network is a 2-layer feed-forward net (640-dim) over the previous-label embedding; monotonic alignments [Tripathi & Lu⁺ 19].
 3. **Full-context transducer** [Graves 12]: LSTM prediction network; monotonic alignments [Tripathi & Lu⁺ 19].
- ▶ Quantization: QAT, per-tensor symmetric, 8 and 4-bit weights, 8-bit activations; zero-point 0.
- ▶ Stages: (1) Full precision baseline → (2) Full Pre-training with QAT (fake quantization) → (3) memristor simulation (3-5 runs).

CTC Experiments

Table: WER (%) on LibriSpeech dev-other - CTC

(a) Baseline

Model	Lex-Free		Tree
	No LM	No LM	4-gram
CTC	7.1	5.0	5.2

(b) QAT

Bits	Lex-Free		Tree
	No LM	No LM	4-gram
8	7.3	5.2	5.3
4	8.3	5.4	5.7

(c) Memristor Simulation

Bits	Lex-Free		Tree
	No LM	No LM	4-gram
8	13.53 ± 0.37	10.35 ± 0.20	6.67 ± 0.09
4	11.85 ± 0.18	TBD	6.77 ± 0.12

FFNN Transducer Experiments: 8-bit Quantization

Table: WER (%) on LibriSpeech dev-other (8-bit Quantization vs. Baseline)

Modules Quantized			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline – Full Precision			5.9	5.8	4.8
QAT					
✓	—	—	6.3	6.1	4.9
✓	✓	—	6.4	6.2	5.0
✓	✓	✓	6.4	6.2	4.9
Memristor Simulation					
✓	—	—	9.76 ± 0.06	8.74 ± 0.03	6.70 ± 0.07
✓	✓	—	10.72 ± 0.10	8.86 ± 0.11	6.67 ± 0.04
✓	✓	✓	TBD	TBD	TBD

FFNN Transducer Experiments: 4-bit Quantization

Table: WER (%) on LibriSpeech dev-other (4-bit Quantization vs. Baseline)

Modules Quantized			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline – Full Precision			5.9	5.8	4.8
QAT					
✓	–	–	7.2	6.8	5.2
✓	✓	–	TBD	TBD	TBD
✓	✓	✓	TBD	TBD	TBD
Memristor Simulation					
✓	–	–	10.09 ± 0.05	9.07 ± 0.11	6.65 ± 0.07
✓	✓	–	TBD	TBD	TBD
✓	✓	✓	TBD	TBD	TBD

FFNN Transducer: Fine-Tuning vs. QAT from Scratch

Table: WER (%) on LibriSpeech dev-other: Checkpoint fine-tuning vs. Full Pre-training (PTN).
Configuration - FFNN Transducer with Encoder under QAT

(a) QAT				
Training	Bits	Lex-Free	Tree	
		No LM	No LM	4-gram
Fine-tuned	8	6.6	6.3	5.0
Full PTN	8	6.5	6.2	5.1
Fine-tuned	4	7.2	7.0	5.1
Full PTN	4	7.2	6.8	5.2

(b) Memristor Simulation				
Training	Bits	Lex-Free	Tree	
		No LM	No LM	4-gram
Fine-tuned	8	8.11 ± 0.06	7.53 ± 0.07	5.81 ± 0.08
Full PTN	8	9.76 ± 0.06	8.74 ± 0.03	6.70 ± 0.07
Fine-tuned**	4	13.09 ± 0.06	11.24 ± 0.10	7.18 ± 0.05
Full PTN	4	10.09 ± 0.05	9.07 ± 0.11	6.65 ± 0.07

**Tentative results, to be updated

→ Fine-tuning from a full-precision checkpoint is more effective than training from scratch, for memristor simulation.

Full Context Transducer Experiments: 8-bit Quantization

Table: WER (%) on LibriSpeech dev-other (8-bit Quantization vs. Baseline)

Modules Quantized			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline – Full Precision			5.8	5.6	4.9
QAT					
✓	–	–	5.9	5.7	5.2
✓	✓	–	6.1	5.9	5.2
✓	✓	✓	6.1	5.9	5.2
Memristor Simulation					
✓	–	–	8.96 ± 0.29	8.06 ± 0.16	7.11 ± 0.07
✓	✓	–	TBD	TBD	TBD
✓	✓	✓	TBD	TBD	TBD

Optimal Search Parameters under Noise

Table: Optimal LM scales for recognitions for FFNN Transducer (Tree + 4-Gram LM search, LibriSpeech dev-other).

	ILM Scale	0.1	0.2	0.3	0.4
	ELM Scale	0.6	0.7	0.5	0.5
Baseline	WER (%)	4.8	4.8	4.9**	4.96
QAT Encoder	QAT	5.2	5.0	5.2**	–
	Memristor	17.42 ± 1.60	11.95 ± 0.74	6.69 ± 0.07	–
QAT Encoder & Predictor	QAT	5.00	4.98	–	5.19
	Memristor	18.67 ± 1.58	12.80 ± 0.55	–	6.68 ± 0.04

ILM: Internal Language Model, **ELM:** External Language Model

→ As the model gets noisier, the optimal LM scale increases to compensate for the distortion.

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Future Work: Error Analysis

► Finetuning:

- ▷ Finetuning from a full precision checkpoint for other configurations to improve WER.

► Error Analysis:

- ▷ Substitutions vs. insertions vs. deletions.
 - Why does lexicon-free decoding degrade faster than tree search?
- ▷ Acoustic posterior distortion and score calibration with external LMs.

► Generalization:

- ▷ Evaluation and training on larger/out-of-domain datasets.

Future Work: Search

► Decoder Parameters:

- ▷ Explore penalties for blank emissions.
- ▷ Tune softmax temperature to sharpen or soften posterior peaks under noise.

► Search Space vs. Model & QAT Level:

- ▷ Current search space is large → hurts Real-Time Factor (RTF).
- ▷ Study the impact of restricting the search space on WER and RTF.
 - How many more hypotheses are needed for on-device vs 4-bit vs. 8-bit vs. Full precision runs?
 - Compare hypothesis expansion across CTC and Transducer variants.

► Search Paradigms:

- ▷ Explore alternative search algorithms and language model combinations (e.g. Lexicon-free with LSTM/Transformer based LM).

Future Work: Robust Training & Countermeasures

► Noise-Aware QAT:

- ▷ Inject simulated crossbar conductance noise directly into training forward pass.

► Selective Layer Mapping:

- ▷ Deal with noise in noise-sensitive components (e.g., joint network).
→ Mixed precision quantization

► Denoising & Output Compensation:

- ▷ Post-processing to correct analog inaccuracies.
→ Train a small neural network to denoise the memristor output.
→ Use a small calibration dataset to learn a linear transformation to correct the output.

Thank you for your attention!

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Appendix: Alternative Table Formats (FFNN Transducer)

Table: WER (%) on LibriSpeech dev-other (8-bit Quantization vs. Baseline)

Quantized Modules	Lex-Free		Tree
	No LM	LSTM	4-gram
Baseline (Full Precision)	–	–	–
QAT			
Encoder	–	–	–
Encoder + Predictor	–	–	–
Encoder + Predictor + Joiner	–	–	–
Memristor Simulation			
Encoder	–	–	–
Encoder + Predictor	–	–	–
Encoder + Predictor + Joiner	–	–	–

Appendix: Alternative Table Formats (Additive Labels)

Table: WER (%) on LibriSpeech dev-other (8-bit Quantization vs. Baseline)

Quantized Modules	Lex-Free		Tree
	No LM	LSTM	4-gram
Baseline (Full Precision)	–	–	–
QAT			
Encoder	–	–	–
+ Predictor	–	–	–
+ Joiner	–	–	–
Memristor Simulation			
Encoder	–	–	–
+ Predictor	–	–	–
+ Joiner	–	–	–

Appendix: Device VMM Operation Statistics

operation	average	std-dev	min	max
1.0×1	1.016	0.063	0.707	1.27
0.1×1	$0.989 \cdot 10^{-1}$	$0.062 \cdot 10^{-1}$	$0.688 \cdot 10^{-1}$	$1.24 \cdot 10^{-1}$
0.01×1	$0.985 \cdot 10^{-2}$	$0.069 \cdot 10^{-2}$	$0.721 \cdot 10^{-2}$	$1.20 \cdot 10^{-2}$
1.0×0	$8.52 \cdot 10^{-4}$	$4.75 \cdot 10^{-2}$	-0.2751	0.2455
1.0×0.5	0.476	0.094	0.109	1.06

Table: Hardware measurement statistics for simulated memristor VMM operations [Rossenbach & Hilmes⁺ 25].